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Problem Set Number: #4 | Due Date: 3/3/26

Acknowledgement: I received help from Alexa Rosano.

Question 1

Known: $t_g = 3\,mm$, $k_g = 1.5\,W/(m \cdot K)$, $t_a = 0.1\,mm$, $k_a = 1\,W/(m \cdot K)$, η, T_{Si},

$\eta = a - b\,T_{Si}$, $a = 0.553$, $b = 0.001\,K^{-1}$, $R_c = 10^{-4}(m^2 \cdot K)/W$, $G = 800\,W/m^2$,

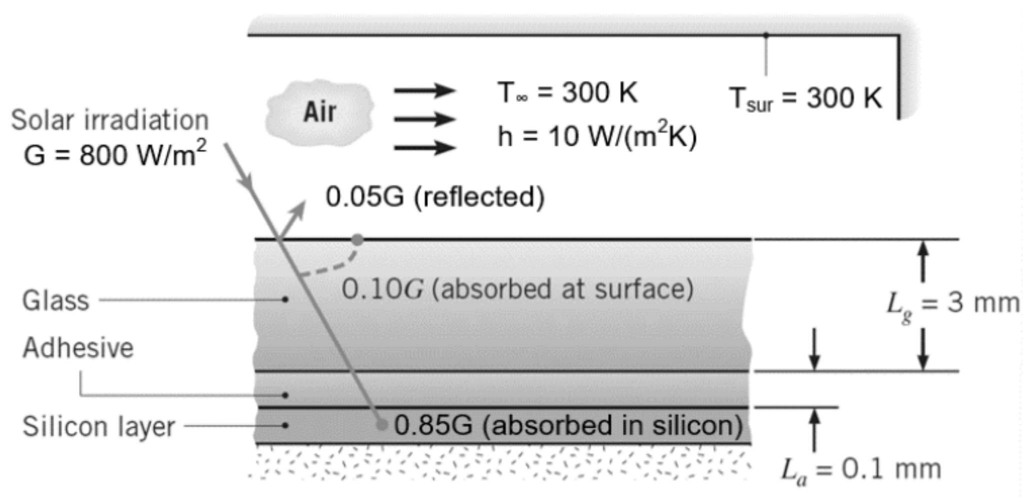
$T_\infty = T_{surr} = 300\,K$, $h = 10\,W/(m^2 \cdot K)$, $h_{rad} = 5\,W/(m^2 \cdot K)$

Find: (a) Show the thermal resistance network that can describe heat transfer across the PV panel. Clearly label all thermal resistances, nodes for temperature, and heat fluxes entering the temperature nodes. State key assumptions that make the thermal resistance network valid. Calculate the value of each thermal resistance in the network.

(b) Calculate the temperatures of the top surface of the glass $T_{g,\,top}$ and the silicon layer T_{Si}.

(c) Calculate the solar-to-electricity conversion efficiency η and resulting electricity power flux produced by the PV panel.

Schematic:



Properties:

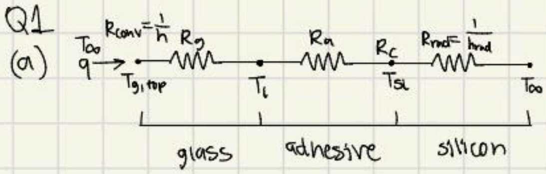
$k_g = 1.5\,W/(m \cdot K)$, $k_a = 1\,W/(m \cdot K)$, $h = 10\,W/(m^2 \cdot K)$, $h_{rad} = 5\,W/(m^2 \cdot K)$

Assumptions: no contact resistance between the glass and the adhesive

Analysis:

Q1

(a)



Key Assumptions:

- No R_c between glass & adhesive
- 1D, steady state heat transfer
- uniform temperature in all regions

$$R_g = \frac{t_g}{k_g} = \frac{0.003\,m}{1.5\,W/m\cdot K} = \boxed{0.002\,\frac{m^2\cdot K}{W}}$$

$$R_{rad} = \frac{1}{h_{rad}}$$

in parallel $\Rightarrow$ $R_{loss} = \frac{1}{h + h_{rad}} = \frac{1}{15\,W/m^2\cdot K} = \boxed{0.0667\,\frac{m^2\cdot K}{W}}$

$$R_a = \frac{t_a}{k_a} = \frac{0.0001\,m}{1\,W/m\cdot K} = \boxed{0.0001\,\frac{m^2\cdot K}{W}}$$

$$R_{conv} = \frac{1}{h}$$

$$\boxed{R_c = 10^{-4}\,m^2\cdot K/W}$$

$$\boxed{R_{cond} = 0.0022\,\frac{m^2\cdot K}{W}}$$

(b) E-Balance: $0.1G + \frac{T_{si} - T_{g,top}}{R_{cond}} = \frac{T_{g,top} - T_\infty}{R_{loss}}$

$\dot{q}''_{elec} = \eta G \rightarrow \dot{q}''_{si,heat} = 0.85G - \eta G \rightarrow \frac{T_{si} - T_{g,top}}{R_{cond}} = 0.85G - \eta G$

$\eta = a - bT_{si} = 0.553 - 0.001\,T_{si}$

$T_{si} = 330\,K \quad , \eta = 0.223$

$\dot{q}''_{si,heat} = (800\,W/m^2)(0.85) - (0.223)(800\,W/m^2) = 502\,W/m^2$

$\Delta T_{cond} = (502\,W/m^2)(2.2\times 10^{-3}\,\frac{m^2\cdot K}{W}) = 1.1\,K \Rightarrow T_{si} \approx T_{g,top} + 1.1\,K$

$\dot{q}'' = (0.1)(800\,W/m^2) + 502\,W/m^2 = 582\,W/m^2$

$\boxed{T_{si} = 340.1\,K}$

$T_{g,top} - 300\,K = (582\,W/m^2)(6.67\times 10^{-2}\,\frac{m^2\cdot K}{W}) \rightarrow \boxed{T_{g,top} \approx 339\,K}$

(c) $\eta = a - bT_i = 0.553 - 0.001\,(340.1\,K) = \boxed{0.213}$

$\dot{q}''_{elec} = \eta G = (0.223)(800\,W/m^2) = \boxed{178.4\,W/m^2}$

Question 2

Known: $D = 10\,mm,\ k = 80\,W/(m \cdot K),\ T_w = 200\,°C,\ L_{ins} = 200\,mm,\ L_0,$

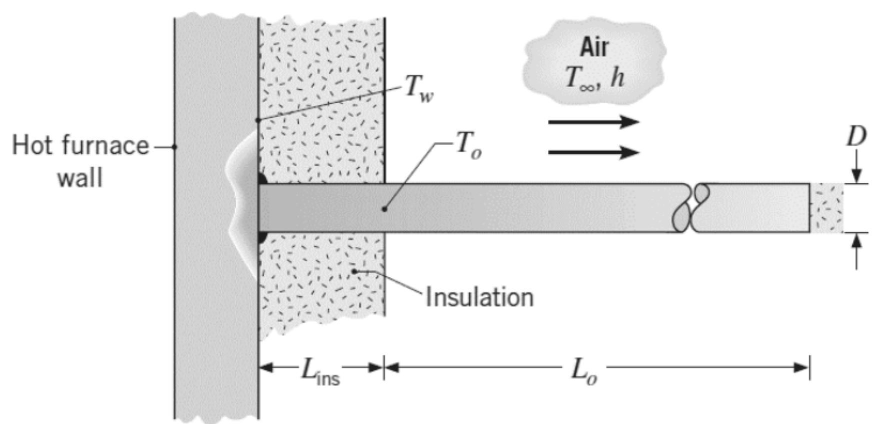
$T_{max} = 100\,°C, T_\infty = 20\,°C, h = 20\,W/(m^2 \cdot K)$

Find: (a) Derive an expression for the exposed surface temperature T_o as a function of the prescribed thermal geometrical parameters.

(b) If exposed length L_o is 200 mm, calculate the corresponding exposed surface temperature T_o. Does the exposed length meet the specified operating limit?

(c) Calculate the minimum exposed length L_o that ensures T_o is below the specified operating limit.

Schematic:



Properties: $k = 80\,W/(m \cdot K),\ h = 20\,W/(m^2 \cdot K)$

Assumptions: the tip is well insulated

Analysis:

(a) $A_c = \frac{\pi D^2}{4} = \frac{\pi(0.01m)^2}{4} = 7.85\times10^{-5} m^2$

$P = \pi D = \pi(0.01m) = 0.03\ m$

$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{(20\ W/m^2{\cdot}K)(0.03\ m)}{(80\ W/m{\cdot}K)(7.85\times10^{-5} m^2)}} = 9.77\ m^{-1}$

Fin Equation: $\frac{T_0 - T_\infty}{T_w - T_\infty} = \frac{\cosh(m(L_0 - x))}{\cosh(mL_0)}$

$T_0 = (T_w - T_\infty)\cdot\frac{\cosh(m(L_0 - x))}{\cosh(mL_0)} + T_\infty \rightarrow \boxed{T_0 = (T_w - T_\infty)\cdot\frac{1}{\cosh(mL_0)} + T_\infty}$

(b) $L_0 = 200\ mm$

$T_0 = (200°C - 20°C)\cdot\frac{1}{\cosh(9.77m^{-1}\cdot 0.2m)} + 20°C$

$T_0 = (180°C)\cdot\frac{1}{\cosh(1.954)} + 20°C \rightarrow \boxed{T_0 = 70°C}$

$T_0 < T_{max}$ so it meets the specified operating limit

(c) $T_0 = (T_w - T_\infty)\cdot\frac{1}{\cosh(mL_0)} + T_\infty$

$\frac{T_0 - T_\infty}{T_w - T_\infty} = \frac{1}{\cosh(mL_0)}$

$\frac{T_w - T_\infty}{T_0 - T_\infty} = \cosh(mL_0)$

$\frac{\cosh^{-1}\left(\frac{T_w - T_\infty}{T_0 - T_\infty}\right)}{m} = L_0 \longrightarrow L_0 = \frac{\cosh^{-1}\left(\frac{200°C - 20°C}{100°C - 20°C}\right)}{9.77m^{-1}} = \frac{\cosh^{-1}(2.25)}{9.77m^{-1}}$

$\boxed{L_0 \approx 0.148\ m}$

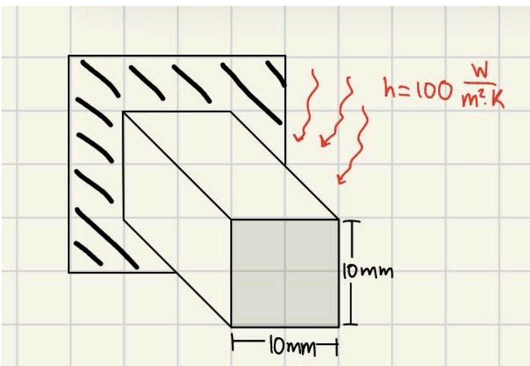
Question 3

Known: $k = 400\, W/(m \cdot K)$, $s = 10\, mm$, $h = 100\, W/(m^2 \cdot K)$, $\eta_f = 0.6$

Find: (a) Calculate the fin length L.

(b) Calculate the thermal resistance of the fin R_f and the fin effectiveness ε_f.

Schematic:



Properties: $h = 100\, W/(m^2 \cdot K)$

Assumptions: the fin has a convection tip

Analysis:

(a) $w,\ h,\ \eta_f,\ k$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{(100\, W/m^2 \cdot K)(0.04\, m)}{(400\, W/m \cdot K)(1\times10^{-4} m^2)}} = 10\, m^{-1}$$

$$\eta_f = \frac{\tanh(mL)}{mL} \rightarrow \eta_f mL = \tanh(mL) \rightarrow \tanh^{-1}(\eta_f mL) = mL \rightarrow \boxed{L = 0.151\, m}$$

(b) $$R_f = \frac{1}{\eta_f h \underset{L_{tL}}{A_f}} = \frac{1}{(0.6)(100\, W/m^2 \cdot K)(0.00614\, m^2)} = \boxed{2.714\, K/W}$$

$$\varepsilon_f = \frac{q_f}{q_{nf}} = \left(\frac{kP}{hA_c}\right)^{\frac{1}{2}} \tanh(mL) = \left(\frac{(400\, W/m \cdot K)(0.04\, m)}{(100\, W/m^2 \cdot K)(1\times10^{-4} m^2)}\right)^{\frac{1}{2}} \cdot \tanh(10\, m^{-1} \cdot 0.151\, m) = \boxed{36.28}$$

$$A_c = (0.01\, m)^2 = 1\times10^{-4}\, m^2$$